1911

The delegate list of the first Solvay Conference included many important scientists including Ernest Rutherford, Max Planck, Albert Einstein, and Marie Curie.

> **"** WE WOULD LIKE TO HOPE THAT **OUR CONFERENCE** WILL… **EXERT** AN IMPORTANT **INFLUENCE** ON THE **DEVELOPMENT OF PHYSICS. "**
>
> **Walther Nernst, Principal instigator of the Solvay Conference,** 1911

NEW ZEALAND-BORN PHYSICIST Ernest Rutherford was gathering evidence for a **new theory of atomic structure**. Experiments had indicated that atoms had a dense core, so the plum pudding model was wrong (see 1904). Rutherford proposed that atoms contained a **central dense mass with a positive charge that was surrounded by electrons**. In 1912, he called this central area the **nucleus**, and noted that the value of its charge was related to its atomic mass. Dutch physicist **Antonius Van den Broek** (1870–1926) suggested that the charge value was equal to an element's atomic number, or position in the periodic table. **Henry Moseley** (see 1913) later proved Van den Broek right.

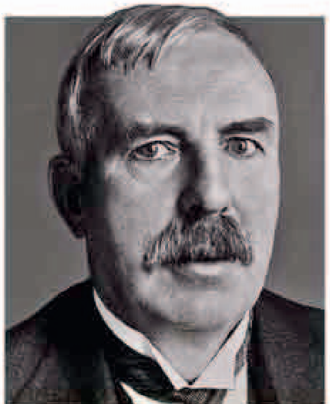
In 1908, Danish physicist **Heike Kamerlingh Onnes** (1853–1926) had liquified helium and was now using it as a coolant to study the electrical properties of frozen mercury. Onnes found that at −452°F (−269°C) mercury's electrical resistance dropped to zero—he had **discovered superconductivity**.

In October, **physicists gathered at the first Solvay Conference**, founded by Belgian industrialist Ernest Solvay. This was the first opportunity **to debate the new field of quantum physics**.

At Columbia University, American biologist Thomas Hunt Morgan (1866–1945) was **experimenting with heredity in fruit flies**. Following on from the work of Gregor Mendel (see 1866) and Hugo de Vries (see 1900), Morgan studied mutations in the insects that caused obvious characteristics, such as eye color, to help him trace patterns of inheritance. In 1911, Morgan discovered that the **genes were on the chromosomes that determined sex**.

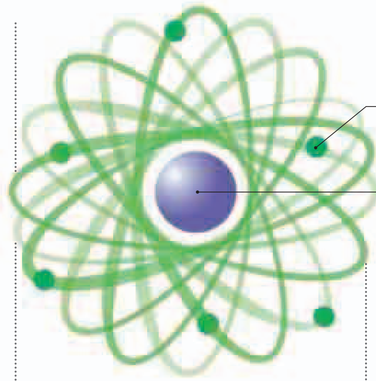
In Antarctica, Robert Scott's (1866–1912) **British Terra Nova expedition set out for the South Pole**. In what was later described as "the worst journey in the world," they succeeded in part of their mission and found a rookery of Emperor Penguins in midwinter. But in December, a Norwegian expedition, led by **Roald Amundsen** (1872–1928), **reached the South Pole first**. The British Polar party all died on their return journey.

German chemist **Philip Monnartz** described a way of **producing steel to improve its corrosion resistance.** The result—**stainless steel**—would be patented in 1912 and in 1913, British engineer Harry Brearley (1871–1948) would cast the first commercial stainless steel in Sheffield, England.



ERNEST RUTHERFORD
(1871–1937)

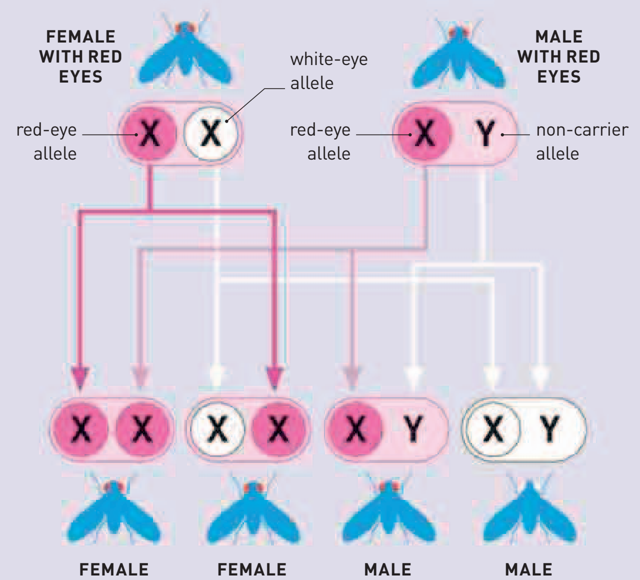
Born in New Zealand, Ernest Rutherford worked at McGill University, Canada, moving to the University of Manchester, England, in 1907. In 1919, he was appointed director of Cambridge's Cavendish Laboratory. Rutherford established the field of nuclear physics with his model of the atom, and explained that radioactivity was caused by atoms decaying into different forms.



Rutherford's atomic model
Rutherford proposed that an atom had a dense nucleus and that the electrons orbited the space around it.

MORGAN'S FRUIT FLY EXPERIMENT

Thomas Hunt Morgan's breeding experiments revealed that the inheritance of a characteristic is linked to gender because its gene occurs on one of the sex chromosomes—usually the X. This can be seen in the eye color of fruit flies because the red-eye variant (allele), is dominant. As a result, in the first generation, a red-eyed female and a white-eyed male would produce only red-eyed offspring, but some will carry the white-eye allele. In the next generation, if a red-eyed female has a white-eye trait it will show up in her sons.



January 20 American **Walter Bradford Cannon** makes **link** between high levels of the hormone **epinephrine and stress**

February 27 American engineer **Charles F. Kettering** demonstrates **first electric starter** for automobiles

March 5 Carl Bosch describes the redesign of the **Haber method** to make ammonia on an industrial scale

March 7 Ernest Rutherford describes atom's dense center—**the nucleus**

April 28 Heike Kamerlingh Onnes discovers superconductivity

May 9 Russians **Boris Rosing** and **Vladimir Zworykin** describe transmitting a scanned image using a **cathode ray tube**

July 20 Antonius Van den Broek shows that **nuclear charge of an element** defines its position in the periodic table

October 29 First **Solvay Conference** meets in Brussels to discuss **radiation and quanta**

November 10 Thomas Hunt Morgan suggests that **genes** reside in chromosomes